NASA SPACE APPS 2026 · FLAME IN FREEFALL

TrustFlame

Fire-safety answers from NASA data, with every sentence checked

[Team name]
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Real video frame, brightened: CFI-G flame on the ISS, NASA PSI (PSI-159)

Fire in space has already cost lives, and the Moon raises the risk

APOLLO 1 · 1967

100%

oxygen in the cabin during a launch-pad test

“During the test, a flash fire swept through the spacecraft so quickly, there was no time to open the hatches.”

NASA Safety and Mission Assurance: Apollo 1

Afterwards: a hatch that opens in three seconds

MIR · 1997

6

crew members aboard when an oxygen generator ignited

“The flame blocked the evacuation route to the Soyuz spacecraft docked at the aft end of Kvant-1”

NASA History: Fire Aboard Space Station Mir

Most likely cause: latex glove material in the canister

MOON BASE · NEXT

≥ 30%

oxygen expected, at 10.2 psi or less (Earth: 21%)

“more than a 40% increase in the oxygen concentration making materials much more flammable.”

NASA Space Apps: Flame in Freefall challenge

NASA's tests of solid materials stop at 22.2%

NASA ran the experiments, but finding the answer is the hard part

24

combustion investigations in NASA's PSI database

2,001

test rows: fuel, oxygen, pressure, air flow

12.3 TiB

of raw video and sensor files in the archive

“the amount of data can be overwhelming. It can be hard to find, harder to compare, and harder yet to interpret.”

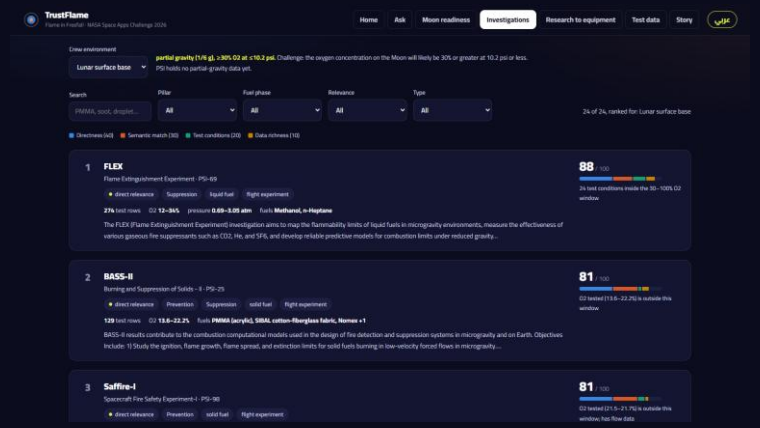
NASA Space Apps: Flame in Freefall challenge

We read every combustion record and every test table, and put oxygen, pressure, flow and material names on one scale.



Real camera frame: BASS test on the ISS, 30 April 2013, NASA PSI (PSI-26)

TrustFlame delivers the three things NASA asked for



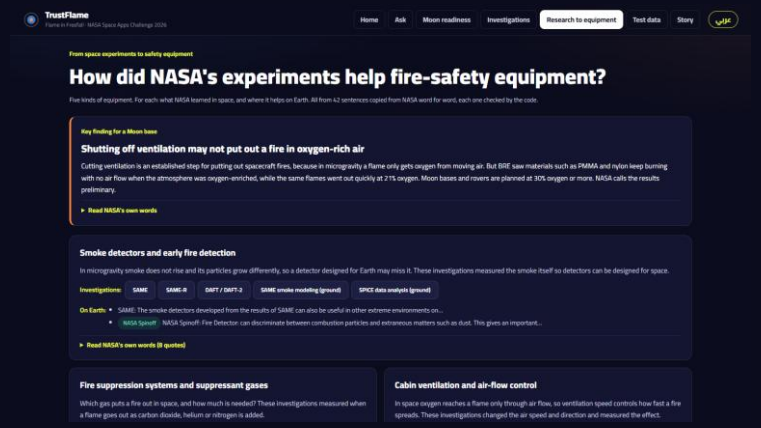
NASA ASKED: IDENTIFY THE MOST RELEVANT INVESTIGATIONS

All 24 investigations ranked 0 to 100 for a station, spacecraft, Moon base, rover or Mars base



NASA ASKED: EXTRACT DATA FOR PREDICTIVE MODELLING

Every NASA test table on one scale, plotted, with CSV export for modellers

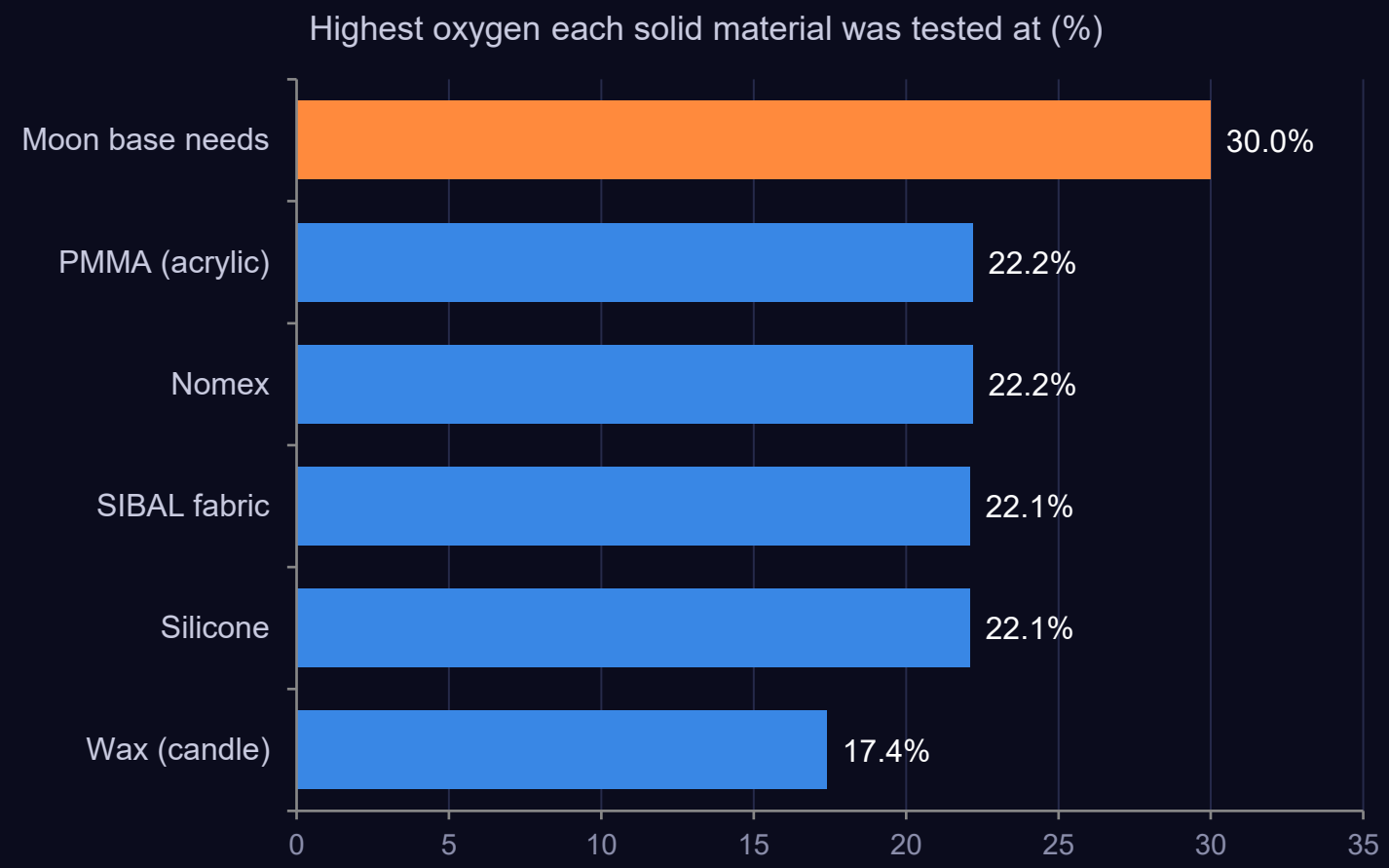


NASA ASKED: HIGHLIGHT RESEARCH FOR FIRE-SAFETY EQUIPMENT

42 NASA quotes, checked word for word, link research to detectors, suppression and materials

Also: a Moon readiness tool · an assistant that takes questions in Arabic or English and answers from NASA sources only · a home page with Ask in front, and a scroll story with real ISS flame video

What we found: no real solid sample was tested above 22.2% oxygen



Computed from the NASA PSI test tables (2,001 rows). Solid materials with recorded oxygen; 5 more from the SAME smoke tests have none recorded.

0

test rows burn a solid material at 30% oxygen or more, or at reduced pressure

THE MISSING TEST

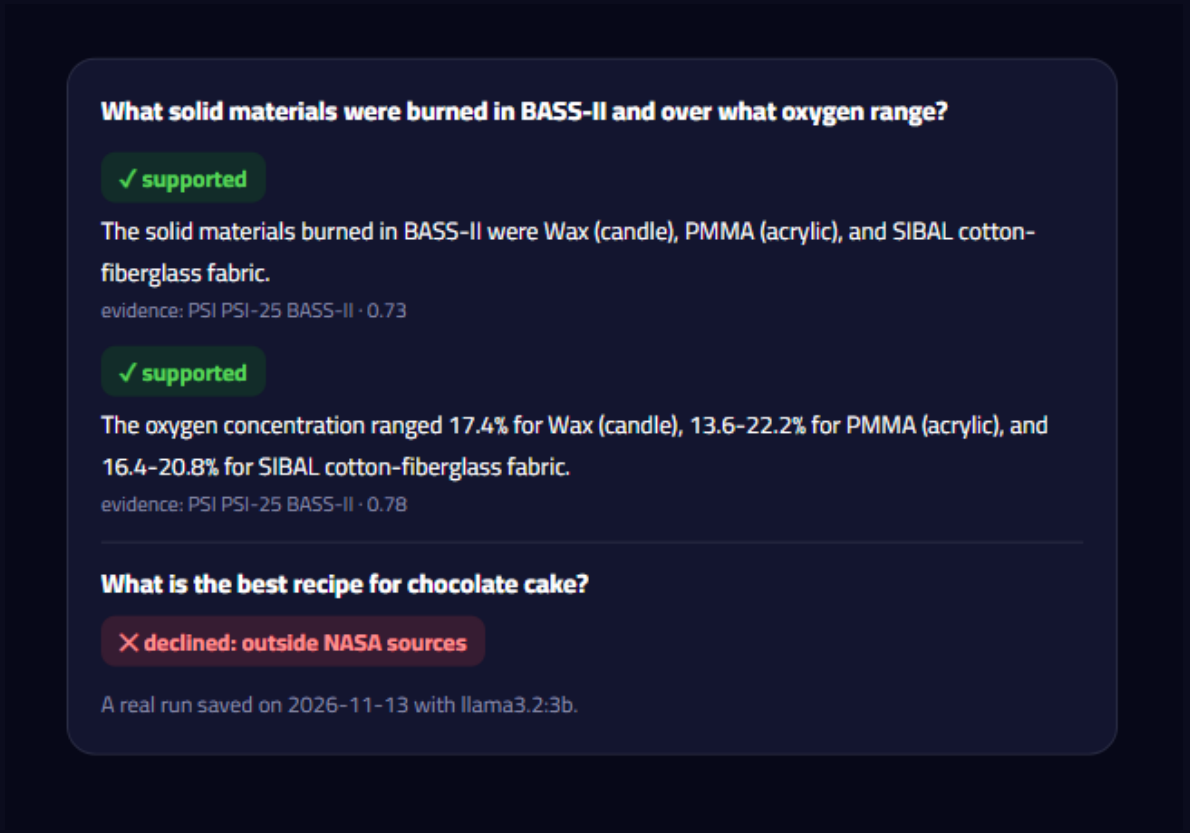
Hardware that tested solids: max 22.2%

Hardware that reached 30%+: gas and liquid only

Test real fabrics and plastics at Moon-base air before building it

How it works: no sentence without evidence, no number without a source

1. **Question** in Arabic or English
2. **Search** 475 passages from NASA sources
3. **Draft** a local model, offline, from the passages only
4. **Guardrail** checks every sentence and every number
5. **Result** verified answer, removed sentence, or refusal



18 / 22 team-written test questions passed

A saved real run of the system, not a mock-up. Limits we state: 1 to 2 minutes per answer on a laptop, the Arabic translation is approximate, PSI has no lunar-gravity data yet.

Impact, and built to last after the hackathon

WHO BENEFITS

- An engineer choosing materials for a Moon base sees in seconds what NASA tested and what it did not
- Arabic-speaking students and researchers read NASA science in their own language

BUILT TO LAST

- No running cost: open models on one laptop, offline
- 6 commands rebuild everything when NASA adds data to PSI
- Open source under Apache-2.0

WHAT COMES NEXT

- A test plan from the gaps: which materials NASA should test first
- Lunar-gravity data (LUCI, FM2) when NASA publishes it
- Check each number against its exact passage

TrustFlame never invents an answer: NASA evidence, or “I don’t know”

Code and data: **github.com/mariadwaghrh/flame-in-freefall** NASA data: PSI Combustion Science and every resource on the challenge page